Predictions and AI for Policy

Lecture 9 – AI4H

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Where are we at?

- Part I: Foundations
  - *Tree-based, deep learning, transformers, LLMs*
- Part II: Policy and ML
  - *Fairness*
  - Prediction policy
  - AI for policy, other uses
- Part III: Causal ML
  - Causal trees and forests
  - Double ML

Overview of this lecture

- Prediction-guided policy
  - Prediction policy cases
    - Medical diagnosis
    - Legal decisions by judges
    - College admission
- LLM-guided policy

Policy vs Prediction

Reminder from last lecture

Prediction vs Policy

- Predicting who will have a heart attack ≠ deciding who gets treatment

- AI systems predict outcomes, but decisions depend on goals and actions
  - Key question: *How can prediction be leveraged for decision?*

- Example domains: healthcare, finance, criminal justice, education

(generalized) Policy problems

- Goal: Choose policy π that maximizes expected utility or welfare for every given x!

- Policy $\pi(x)$: maps input/state, x, to an action/policy, a.

- Optimal policy: $\pi^*(x) \; = \mathrm{argmax}_\pi E[u(y(a))|x]$

  - Again, requires causal understanding how actions influence outcomes
    - estimate **heterogeneous treatment effect**, Y(a|x)
    - > next lecture!

Kleinberg et al. (2015, AER)

- Assume now we have generalized functional form, $\mathrm{u}(\mathrm{y}, \mathrm{a})$
  - Action has two effects on utility, direct and through outcome y
- The dual effect (from total differentiation):

$$\frac{d\mathrm{u}(\mathrm{a}, \mathrm{y})}{d\mathrm{a}} = \frac{\partial \mathrm{u}(\cdot)}{\partial \mathrm{a}} \mathrm{y} + \frac{\partial \mathrm{u}(\cdot)}{\partial \mathrm{y}} \frac{\partial \mathrm{y}}{\partial \mathrm{a}}$$

  - Direct effect $\frac{\partial \mathrm{u}}{\partial \mathrm{a}}$ depends on outcome y
    - A pure prediction problem!
  - Outcome effect ($\frac{\partial \mathrm{y}}{\partial \mathrm{x}}$), also affects utility indirectly ($\frac{\partial \mathrm{u}(\cdot)}{\partial \mathrm{y}}$)
    - Again requires causal problem
- Example – bring rain coat or host garden party

Kleinberg et al. (2018)

- Pre-trial decision by judges
- Solved problem of selective labels – potential unobserved selection
  - Leniency design - verified random
  - Reranking policy

TABLE III

DOES JAILING ADDITIONAL DEFENDANTS BY PREDICTED RISK IMPROVE ON JUDGES? CONTRACTION OF THE MOST LENIENT JUDGES' RELEASED SET

	Judges Relative to most lenient quintile		Algorithm To achieve judge's	
	Δ Jail	Δ Crime	Δ Crime Δ Jail	Δ Jail Δ Crime
Second quintile	0.066	−0.099	0.028	−0.201
Third quintile	0.096	−0.137	0.042	−0.269
Fourth quintile	0.135	−0.206	0.068	−0.349
Fifth quintile	0.223	−0.307	0.112	−0.498

Notes. This table reports the results of contrasting the cases detained by the second through fifth most lenient quintile judges compared with the most lenient quintile judges, and to a release rule that detains additional defendants in descending order of predicted risk from an algorithm trained on failure to appear. The first column shows from where in the predicted risk distribution each less lenient quintile's judges could have drawn their marginal detainees to get from the most lenient quintile's release rate down to their own release rate if judges were detaining in descending order of risk. The second column shows what share of their marginal detainees actually come from that part of the risk distribution. The third column shows the increase in the jail rate that would be required to reach each quintile's reduction in crime rate if we jailed in descending order of the algorithm's predicted risk, while the final column shows the reduction in crime that could be achieved if we increased the jail rate by as much as the judge quintile shown in that row.

Prediction policy cases

Medical diagnosis

Setup

- We want to think about situations where we separate
  - medical diagnosis (outcome)
  - medical treatment (action)
- AI can be used for both
  - Focus here on *diagnosis* – prediction problem
  - Whether to give medical treatment – can use causal prediction methods
    - (part III of course)

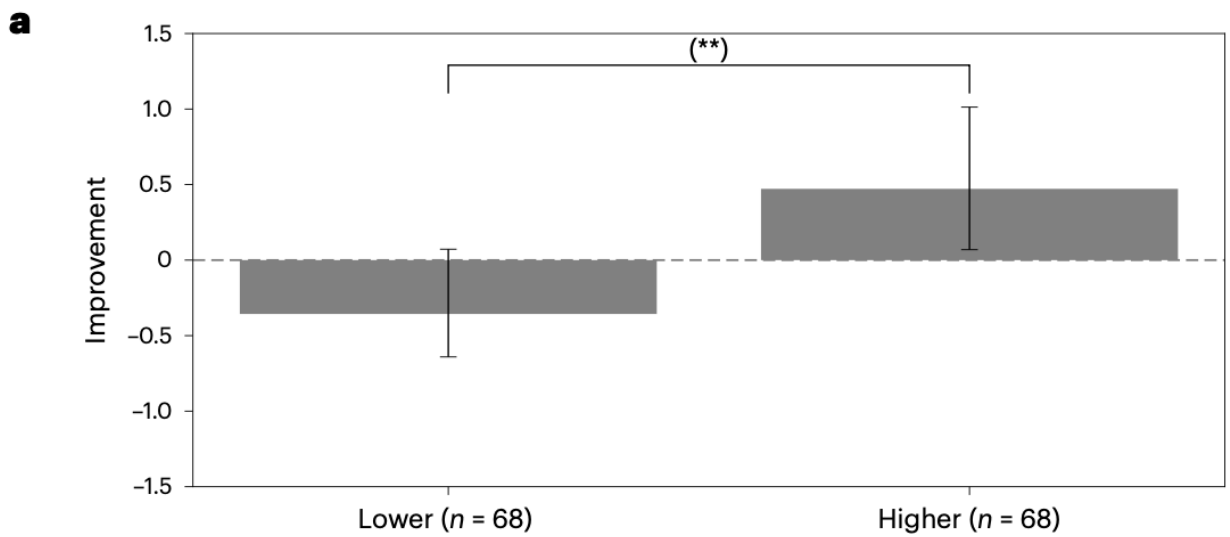
Yu et al. (2024, Nat. Med.)

- Setup
  - 60 patient cases, radiologists examined
    - half without clinical histories
    - half with clinical histories
  - Among these random exposure to AI assistance

- Main outcome – correct diagnosis
  - measured as predicted diagnosis same as ground truth (yes vs. no)

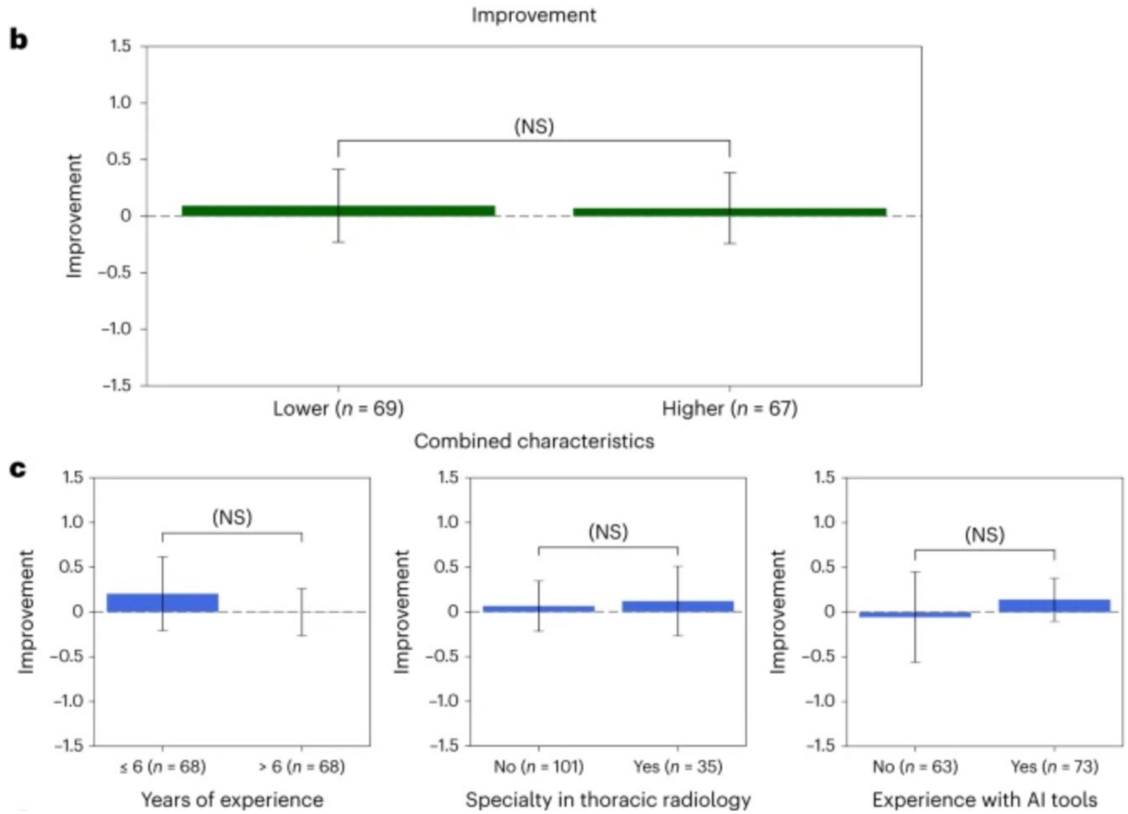
Heterogeneous impact

- Treatment effect of AI
  - Compute individual correctness with and without AI



Origin of heterogeneity - doctor characteristic

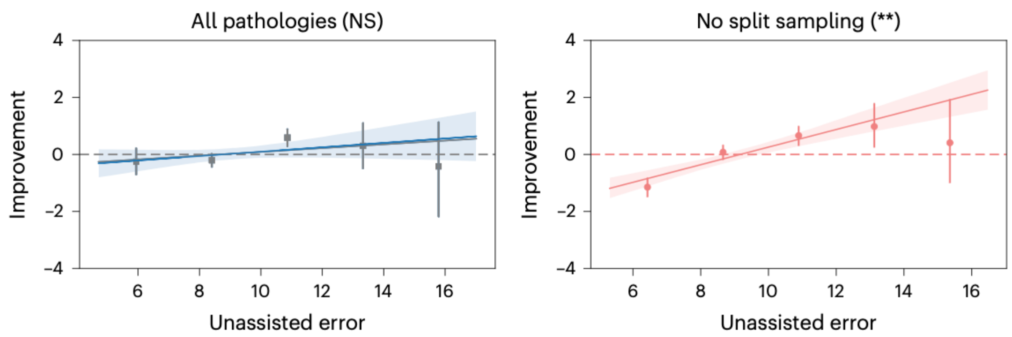
- Driven by doctor backgrounds?
  - years of experience, subspecialty and AI experience not predictive



Origin of heterogeneity - skill

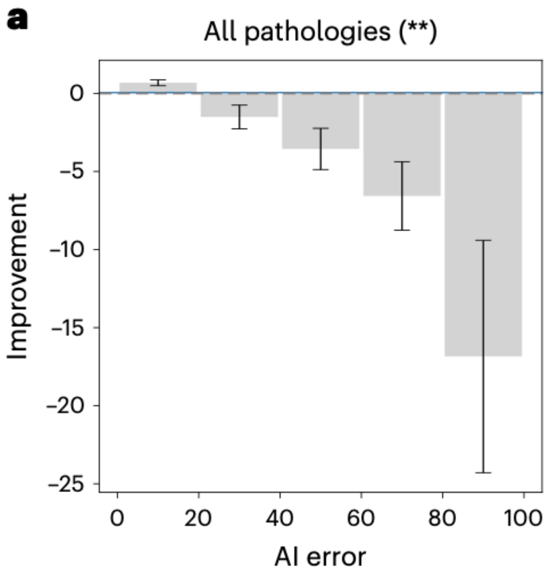
- Driven by doctor latent skill?
  - Measure as unassisisted performance

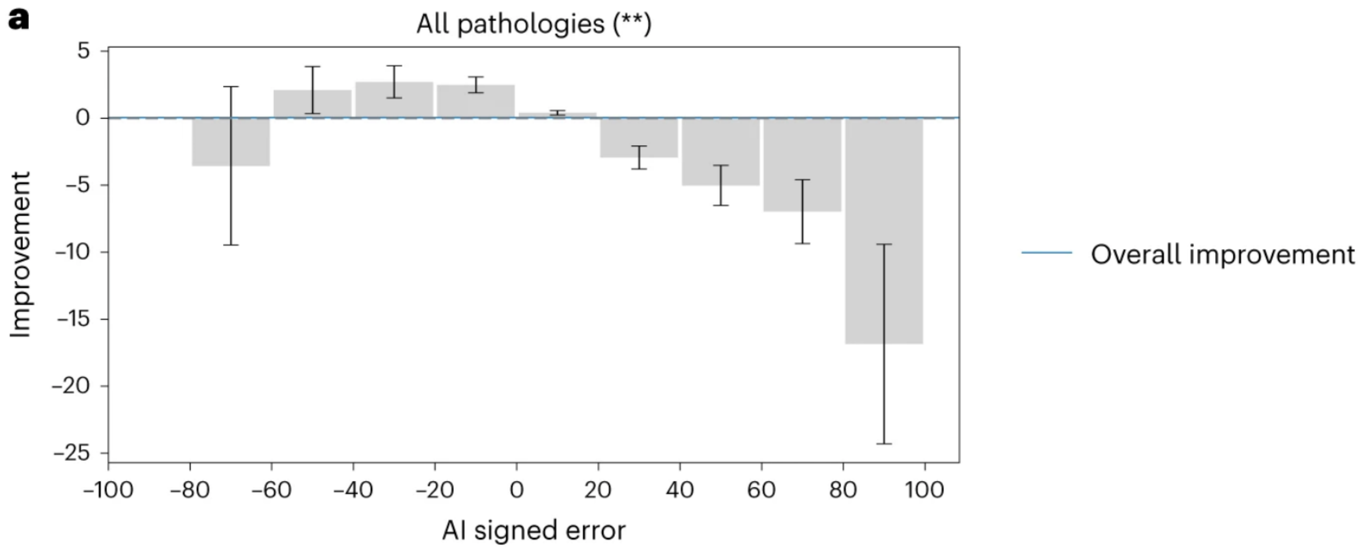
a



Origin of heterogeneity – AI skill

- Driven by AI accuracy?
  - AI error: $|y - \hat{y}_{\mathrm{AI}}|$
  - AI signed error: $y - \hat{y}_{\mathrm{AI}}$





Summary

- Yu et al. find
  - Null effect of impact on AI on decision quality
  - Large heterogeneity in impact of AI!

  - Mainly driven by AI accuracy

- Now – dive into Agarwal et al.

More about the study

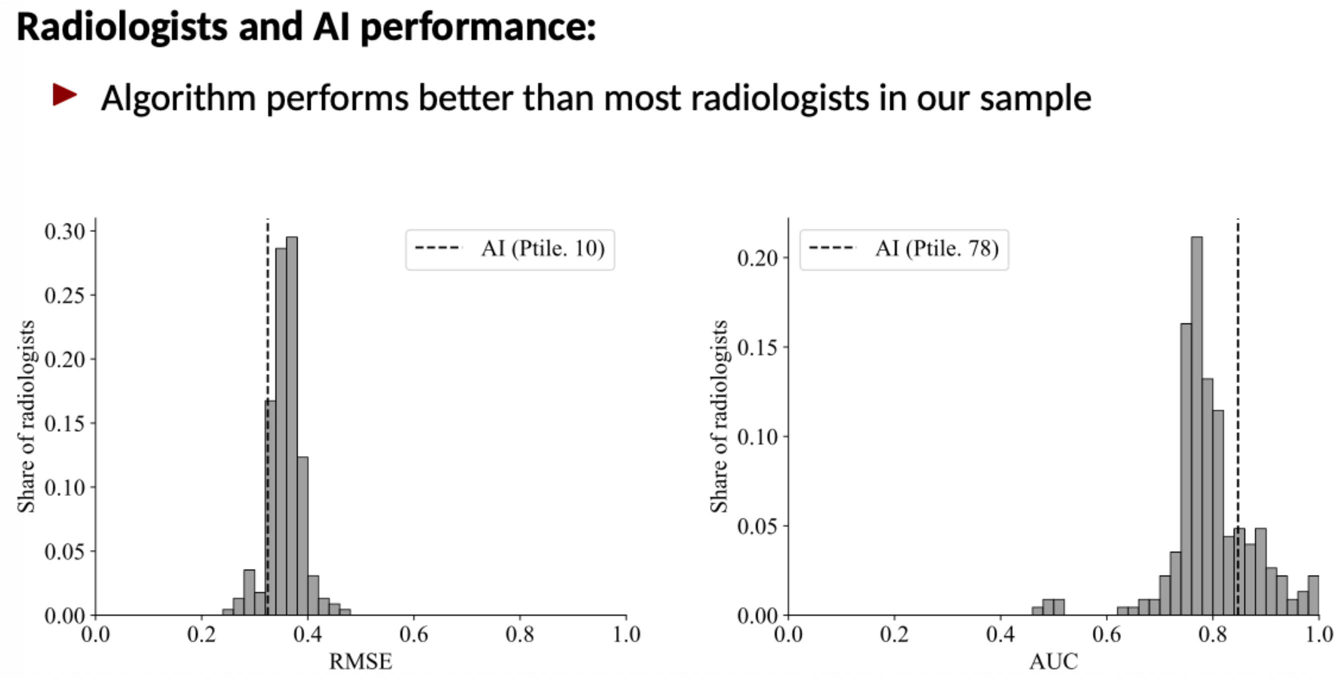
- Underlying data can be published in *Scientific Data* (here)

- Two more more papers

  - Agarwal, Moehring, Rajpurkar & Salz (2024), “Combining human expertise with artificial intelligence: Experimental evidence from radiology”
  - Agarwal, Huang, Moehring, Rajpurkar, Salz & Yu (2024, AEA P&P) “Comparative advantage of humans versus AI in the long tail”

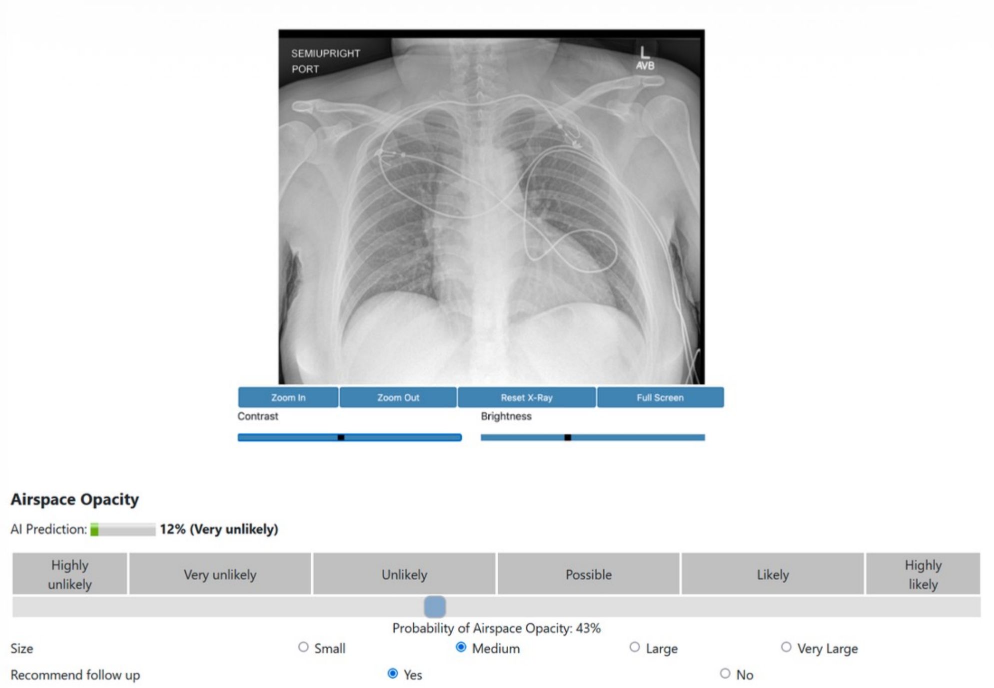
Agarwal et al. (2024)

- Same data as Yu et al.
- New question – how do subjects update their assessment?



Treatment dimensions

Arm 1 – clinical history



Arm 2 – clinical history

Indication

30 years of age, Female, history of hypertension, abnormal EKG, abdominal pain, evaluate for cardiomegaly or mediastinal widening.

Vitals

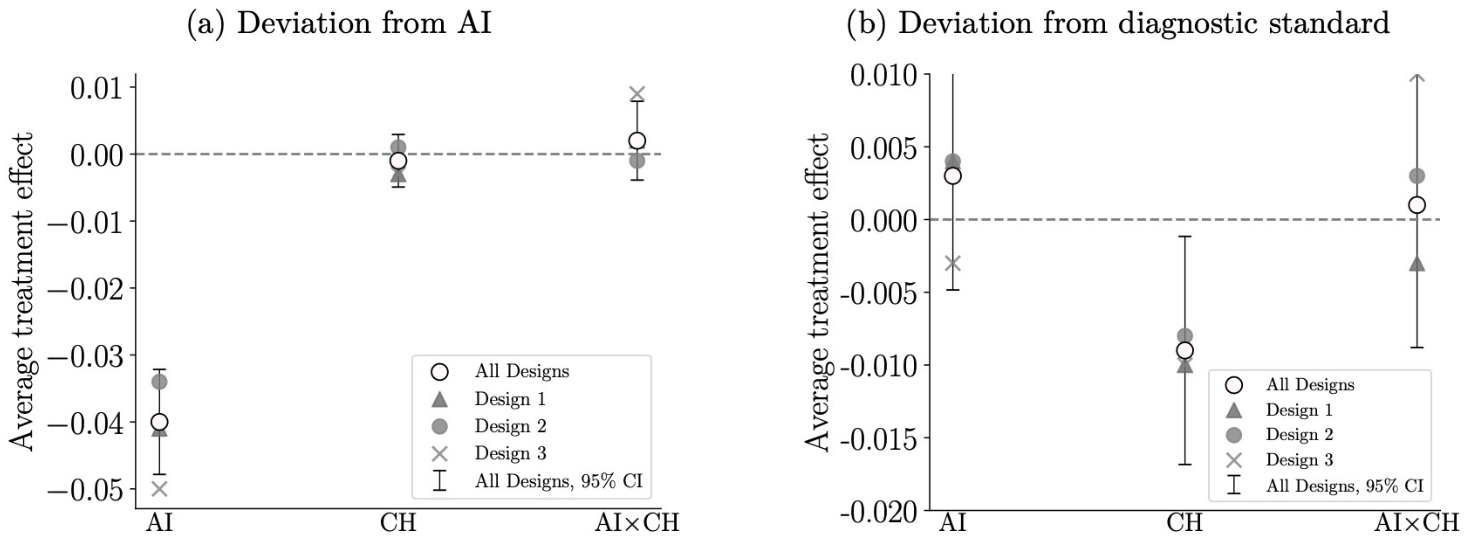
Variable	**Value**
Weight	170 lbs
BP	243/166 mmHg
Temp	99.1F
Pulse	99.0 bpm
Age	30

Abnormal Labs All Labs

Variable	**Value**	**Unit**	**Flag**
ALT (SGPT), Ser/Plas	38.0	U/L	High
AST (SGOT), Ser/Plas	39.0	U/L	High
Eosinophil, Absolute	0.01	K/uL	Low

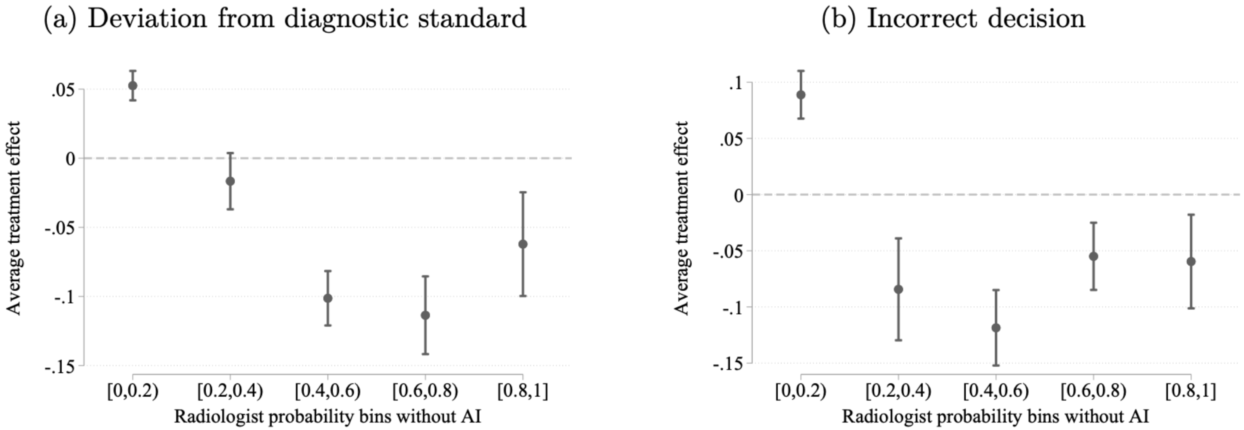
Impact of treatments

Figure 2: Treatment effects of informational interventions



Note: ATE estimated using equation (4), on (a) the deviation from AI and (b) the deviation from the diagnostic standard. Results are for the two top-level pathologies with AI predictions, airspace opacity and cardiomediastinal abnormality; separated by design and pooled across all designs. Standard errors are two-way clustered at the radiologist and patient level.

AI impact by radiologist predictions



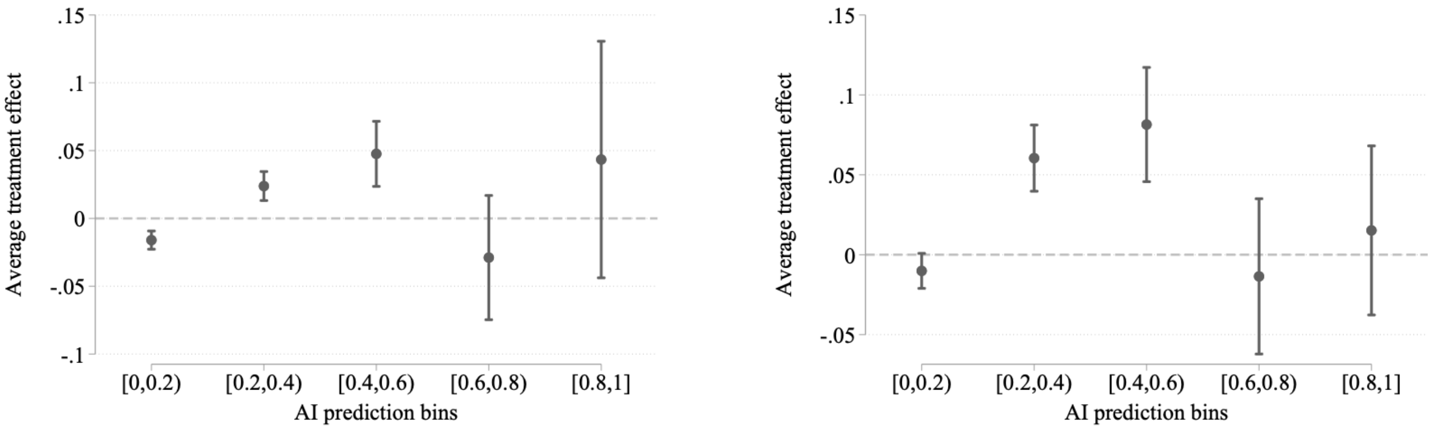
Note: Panel (a) shows the conditional ATE of providing AI information on the deviation from diagnostic standard. Panel (b) shows analogous treatment effects on incorrect diagnosis. Estimates from the regression $Y_{iht} = \gamma_{g_i} + \gamma_{AI} \cdot d_{AI}(t) + \varepsilon_{iht}$ estimated separately for bins of the radiologists' reported probability without AI. Standard errors are two-way clustered at the radiologist and patient level, with 95% confidence intervals depicted. Robustness to experimental design is in appendix C.5.

- Notes:
  - Diagnostic standard (binary)
  - Incorrect decision (probability)

AI impact by AI predictions

Figure 4: Effect of AI by AI prediction

(a) Deviation from diagnostic standard (b) Incorrect decision



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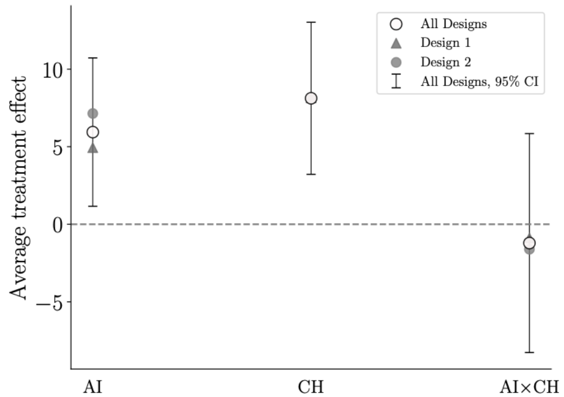
Impact on time use (seconds)

Figure 5: Effect of informational interventions on time



Note: ATE of treatments, estimated using equation (4), on effort measured in terms of active time (in seconds). Results exclude five observations with missing time measures, and observations from design three because of potential order effects. Active time is winsorized to the 95th percentile. Standard errors are two-way clustered at the radiologist and patient level.

Conceptual framework

- A mode for human – AI classification problems

- Suppose there is a human decision-maker, h, faced with case i.
  - She/he must take a binary action ($a_{i,h} \in \{0,1\}$)
  - There is an unobserved binary state ($\omega_i \in \{0,1\}$)
    - > must be predicted
  - The realized payoff depends both on action and ground truth, $u_h\left(a_{i,h}, \omega_i\right)$

- The human can get signals of
  - AI-provided: $s_i^A \in S_A$
  - Own interpretation: $s_{ih}^H \in S_H$
  - Denote human belief $p_h(\omega|s_{ih})$, where $s_{ih} \subset \{s_i^A, s_{ih}^H\}$
    - >They do not necesarrily observe s_i^A

Conceptual framework – optimal action

- Assume human goal is to correct classification
- $u_h(a, \omega) = -1_{a=1,\omega=0} \cdot c_{FP,h} - 1_{a=0,\omega=1} \cdot c_{FN,h}$
  - First component : disutility from false positives
  - Second component: disutility from false negatives

- Optimal action

$$a_h^*(s; p_h) = 1 \cdot \left\{ \frac{p_h(\omega = 1|s)}{p_h(\omega = 0|s)} > c_{rel,h} \equiv \frac{c_{FP,h}}{c_{FN,h}} \right\}.$$

  - Intuition > trades off errors by their relative perceived cost

Belief updating

- Optimal action can is based on optimal weighting of signals (Bayesian updating). Optimal updating implies

$$\log \frac{\pi_h\left(\omega_i = 1 \,\middle|\, s_i^A, s_{ih}^H\right)}{\pi_h\left(\omega_i = 0 \,\middle|\, s_i^A, s_{ih}^H\right)} = \log \frac{\pi_h\left(s_i^A \,\middle|\, \omega_i = 1, s_{ih}^H\right)}{\pi_h\left(s_i^A \,\middle|\, \omega_i = 0, s_{ih}^H\right)} + \log \frac{\pi_h\left(\omega_i = 1 \,\middle|\, s_{ih}^H\right)}{\pi_h\left(\omega_i = 0 \,\middle|\, s_{ih}^H\right)}, \tag{3}$$

- Are doctors updating correctly? If not either under- or overweighting AI?

Testing belief updating - bias

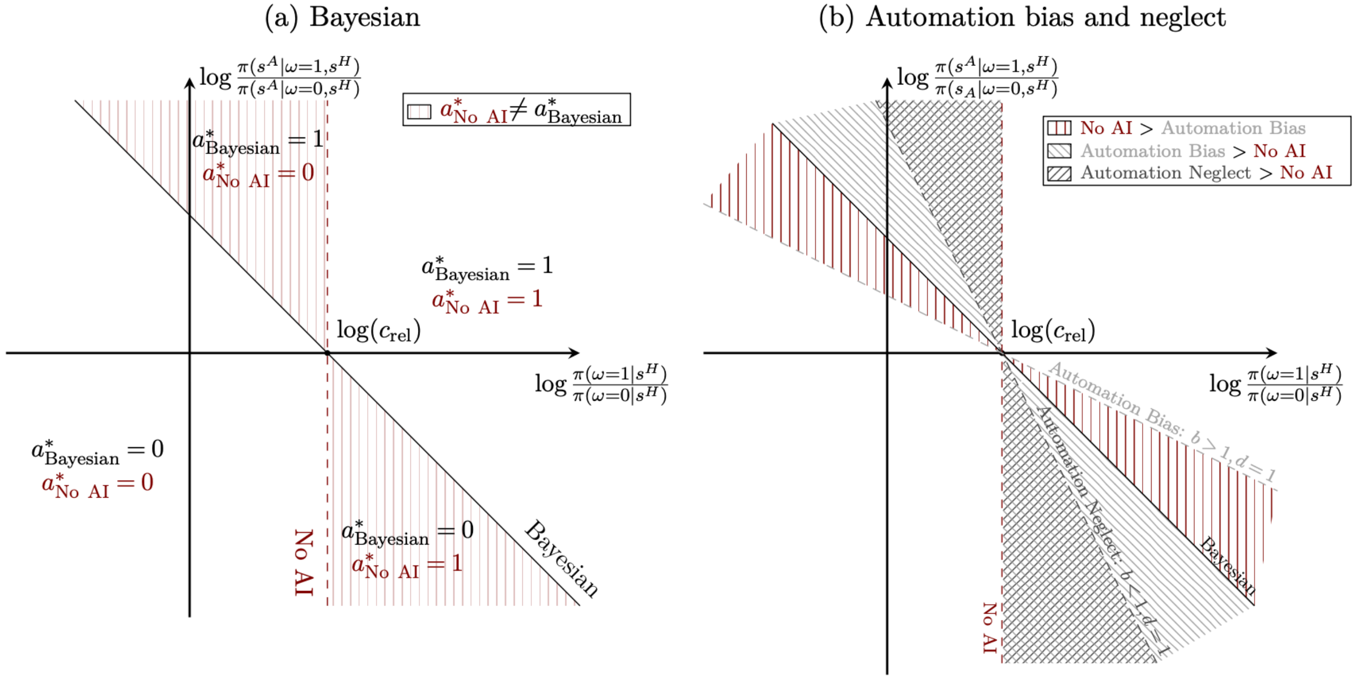
- We can assess consequences of belief updating

$$\log \frac{p_h\left(\omega_i = 1 | s_i^A, s_{ih}^H\right)}{p_h\left(\omega_i = 0 | s_i^A, s_{ih}^H\right)} = b_h \log \frac{\pi_h\left(s_i^A | \omega_i = 1, s_{ih}^H\right)}{\pi_h\left(s_i^A | \omega_i = 0, s_{ih}^H\right)} + d_h \log \frac{\pi_h\left(\omega_i = 1 | s_{ih}^H\right)}{\pi_h\left(\omega_i = 0 | s_{ih}^H\right)}, \qquad (5)$$

  - Interpretation:
    - $b_h > d_h$ - automation bias
    - $b_h < d_h$ - automation neglect
  - Consequence: suboptimal decisions!

Visualizing biased updating

Figure 6: Comparing decisions with and without AI assistance



Note: (a) Decision criterion of a Bayesian with and without AI assistance and where their decisions align. Shaded regions show the regions in which AI improves or worsens decision making. (b) Where the decisions of an expert as a function of the signals disagree with a Bayesian in cases with and without AI assistance in the presence of automation bias or neglect when $d = 1$.

Testing belief updating - bias

- We can assess consequences of belief updating

$$\log \frac{p_h\left(\omega_i = 1 | s_i^A, s_{ih}^H\right)}{p_h\left(\omega_i = 0 | s_i^A, s_{ih}^H\right)} = b_h \log \frac{\pi_h\left(s_i^A | \omega_i = 1, s_{ih}^H\right)}{\pi_h\left(s_i^A | \omega_i = 0, s_{ih}^H\right)} + d_h \log \frac{\pi_h\left(\omega_i = 1 | s_{ih}^H\right)}{\pi_h\left(\omega_i = 0 | s_{ih}^H\right)}, \tag{5}$$

  - Interpretation:
    - $b_h > d_h$ - automation bias
    - $b_h < d_h$ - automation neglect
  - Consequence: suboptimal decisions! (Proposition 1)

- Empirical estimation

$$\log \frac{p_h\left(\omega_i = 1 | s_i^A, s_{ih}^H\right)}{p_h\left(\omega_i = 0 | s_i^A, s_{ih}^H\right)} = a + b \cdot \log \frac{\pi_h\left(s_i^A | \omega_i = 1, s_{ih}^H\right)}{\pi_h\left(s_i^A | \omega_i = 0, s_{ih}^H\right)} + d \cdot \log \frac{\pi_h\left(\omega_i = 1 | s_{ih}^H\right)}{\pi_h\left(\omega_i = 0 | s_{ih}^H\right)} + \varepsilon_{ih}, \tag{7}$$

Testing belief updating – dependence neglect

- Can also estimate

$$\log \frac{p_h\left(\omega_i = 1 | s_i^A, s_{ih}^H\right)}{p_h\left(\omega_i = 0 | s_i^A, s_{ih}^H\right)} = b_h \log \frac{\pi_h\left(s_i^A | \omega_i = 1\right)}{\pi_h\left(s_i^A | \omega_i = 0\right)} + d_h \log \frac{\pi_h\left(\omega_i = 1 | s_{ih}^H\right)}{\pi_h\left(\omega_i = 0 | s_{ih}^H\right)}, \tag{6}$$

  - Idea – people may not factor in own signal’s correlation with AI
    - > comes from the same material!
  - Consequence: suboptimal decisions! (Proposition 2)

Estimation challenges

- We need to unpack ratio of conditional densities

$$\log \frac{\pi_h\left(s_i^A | \omega_i = 1, s_{ih}^H\right)}{\pi_h\left(s_i^A | \omega_i = 0, s_{ih}^H\right)} = \log \frac{\pi_h\left(\omega_i = 1 | s_i^A, s_{ih}^H\right)}{\pi_h\left(\omega_i = 0 | s_i^A, s_{ih}^H\right)} - \log \frac{\pi_h\left(\omega_i = 1 | s_{ih}^H\right)}{\pi_h\left(\omega_i = 0 | s_{ih}^H\right)}.$$

- But human assessment is unobserved
    - Impute/predict from non-AI observations
    - Use instrument of observations from other participants not observing AI

Estimation results

- Major findings:
  - Automation neglect
  - Signal dependence neglect
    - (best data fit)

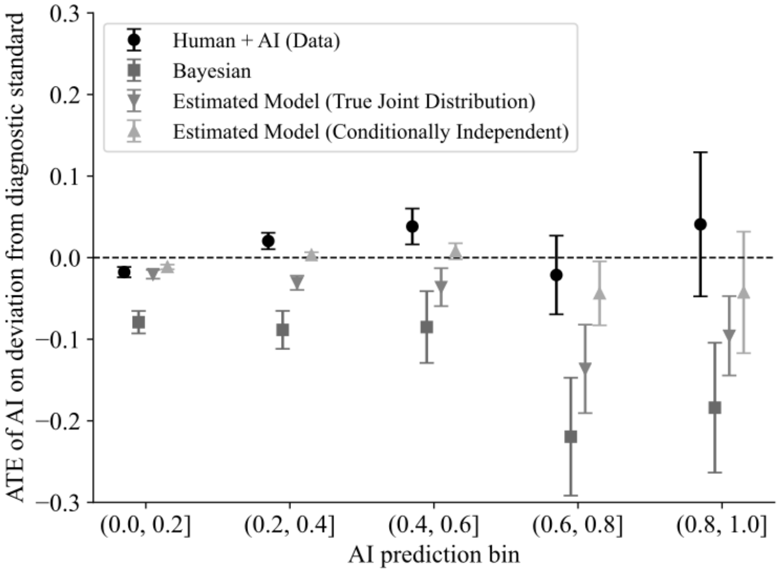
Table 2: Selecting between models of belief updating: top level pathologies with AI

	(1)	(2)	(3)	(4)	(5)	(6)
Automation bias (b)	0.27	0.33	0.12	0.19	0.21	0.12
	(0.02)	(0.03)	(0.02)	(0.03)	(0.03)	(0.02)
Own information bias (d)	1.11	1.09	1.05	1.07	1.07	1.05
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Constant	0.39	0.39	0.25	0.32	0.32	0.25
	(0.04)	(0.04)	(0.03)	(0.03)	(0.04)	(0.03)
No signal dependence	✓			✓		
No pathology dependence	✓	✓		✓	✓	
No clinical history dependence	✓	✓	✓			
Correct updating						✓
J-Statistic	13.08	11.63	8.85	7.53	8.55	7.72
MMSC-BIC	-29.11	-26.34	-8.03	-13.57	-12.55	-9.16
Selected moments	13	12	7	8	8	7
Possible moments	14	14	14	14	14	14
Observations	11420	11420	11420	11420	11420	11420
R-Squared	0.49	0.48	0.42	0.45	0.45	0.42

Note: Estimates of b and d for different specifications of the update term. The models differ by whether the update term conditions on the signal s_H of the pathology at hand, the AI and the radiologist signals for other pathologies, and the information provided in the clinical history of a patient. Each model is estimated via GMM. The reported MMSC-BIC statistic adjusts the J-statistic for the number of included parameters and moments, awarding bonus terms for models with fewer parameters and fewer rejected moments (see Andrews and Lu (2001) for details). The full set of models in the selection procedure are presented in table C.13. The update term is estimated via random forest as described in appendix section C.6.2. Standard errors are clustered at the radiologist level. This table uses data from designs 2 and 3 where we observe the same human's assessment of each patient-pathology both with and without AI assistance.

Estimated impact

Figure 7: Data versus model implied treatment effects



Education and child protection

Research in algorithmic policy

- State-of-the-art
  - Comparison of machine vs. human
    - E.g., Kleinberg et al. (2018, QJE) or Rambachan (2024, QJE)
    - Compare human decisions with counterfactual machine decisions

- Open questions
  - Are (standard) prediction algorithms powerful enough for complex data?
    1. Pre-aggregate data into tabular format – apply standard machine learning
    2. Improved policy using ChatGPT or similar model?

  - Could such predictive systems be integrated in allocative mechanisms, e.g., for centralized admission?

An elusive target

Answering the open questions require:

- An outcome of high policy relevance:
  - Outcome can be controlled by policymaker
  - State of outcome matters - monetary or non-monetary consequences

- Granular predictions of humans and/or current policies

- Hard prediction problem of sufficient scale
  - Standard algorithms cannot handle sequential, high dimenstional input
  - Large scale data set for training algorithm

Introduction

We ask

- Can machine learning *outcompete current human-based policies and improve fairness*?
- Can new ML models (transformer in particular) *outperform standard models with non-sequential models* in prediction policy scenarios?
- What about the ethics of data processing – is more data better?

Danish higher educ. admission: policy relevance & complexity

- Centralized admission – admission criteria determines ranking
  - Primary quota (~85%): admission by high-school GPA as criteria
  - Secondary quota (~15%): faculty assessment as criteria
  - Rankings in quotas can be replaced with machine ranking

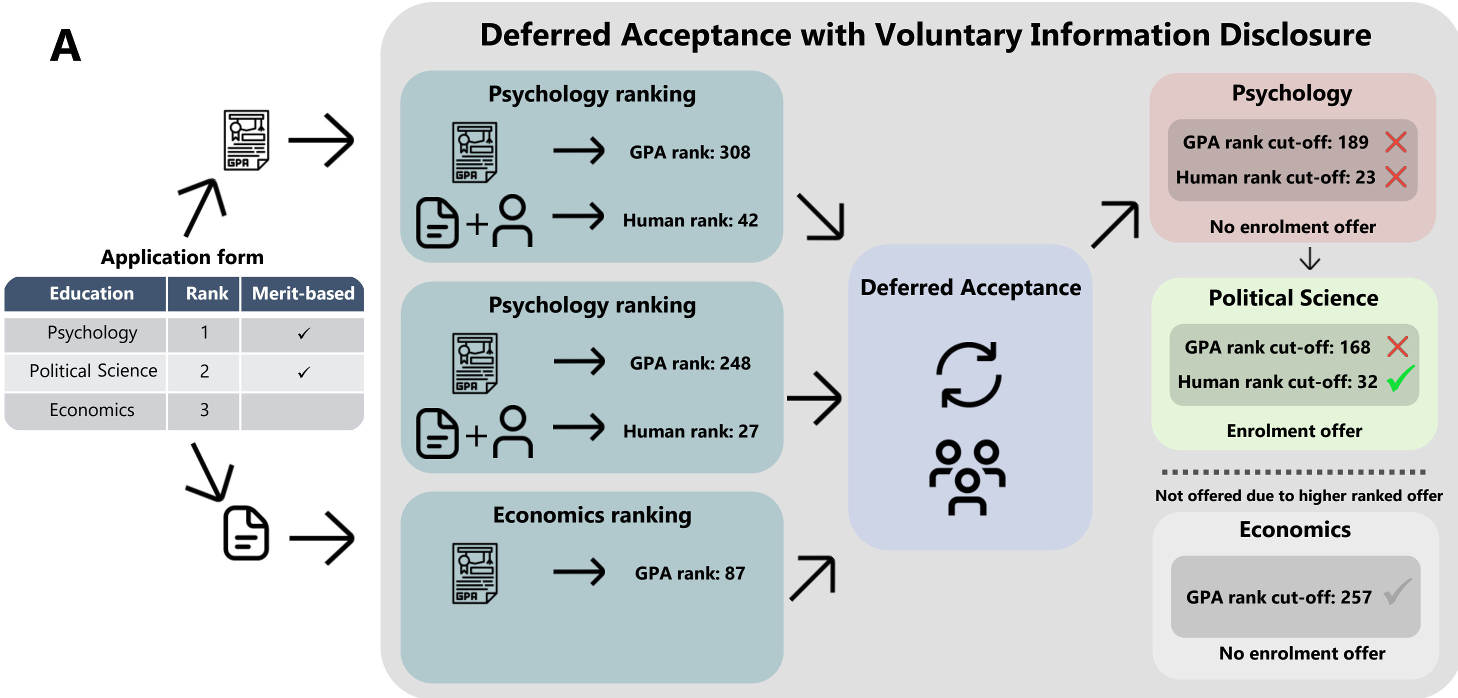
- High stakes – students waste time, programs money (approx. $50,000 per student completed)

- Hard prediction task: no signal in grade transcripts beyond point average

- Registry data – historical data of high quality and large scale

Danish higher educ. admission: allocation system

- Admissions work using Deferred Acceptance
  - Implication: admit those with highest *eligibility*
  - Can view *eligibility* as predicted "risk score"



The methodological problem at hand

Data covers all applicants to higher education in the period 2006-2017
- Grades for primary and secondary school
- Applications to higher education

Repeated observations of the same student
- Hard to encode

Transformers
- Becoming a steady part of machine learning methods and everyday life
- Originally architecture is made for sentence inputs

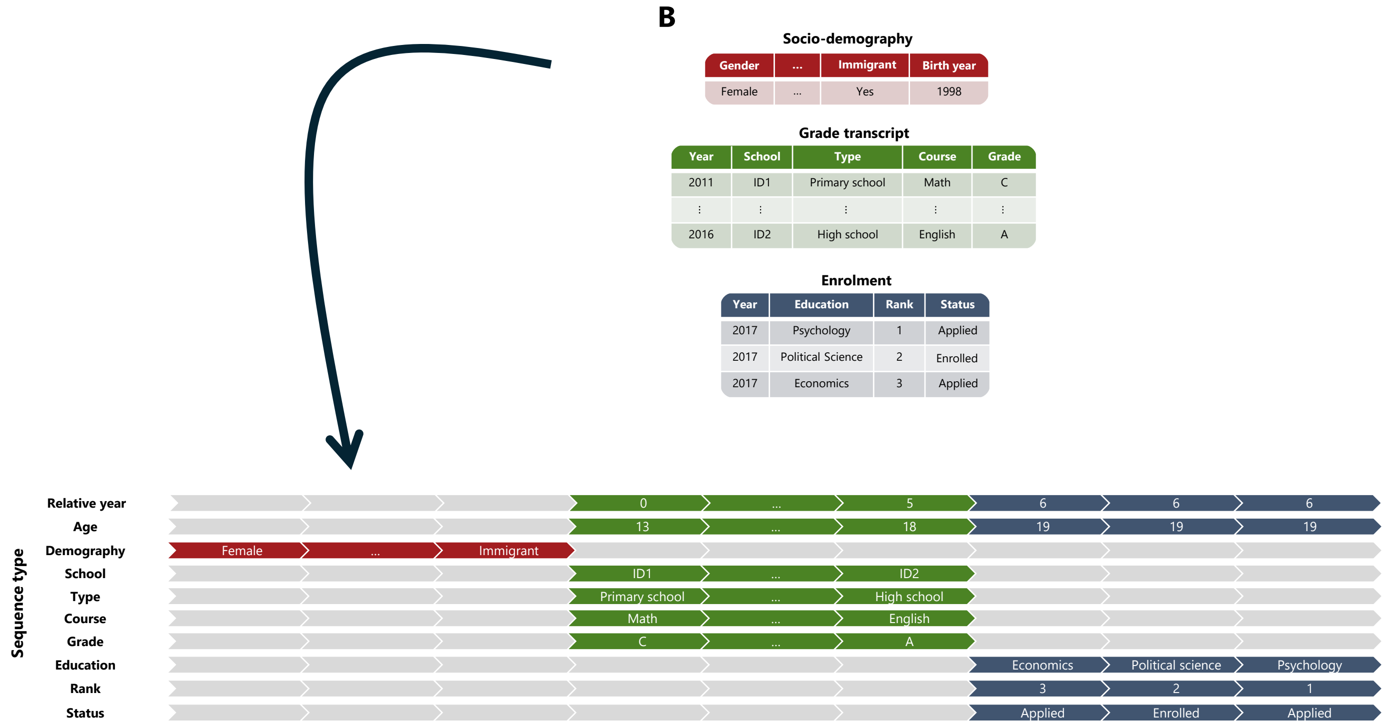
Input and sequences of events

Input issues

- Unknown amount of grades
- Differences in courses

Similar to sentences

- Unknown sentence length
- Heterogeneity in word usage



Two types of models

We create three different input types

- Baseline: only GPA and program
- Academic: enrolment place and transcripts
- All: Full application and background information

Why create different models?

- Separate gains from models and information sets (input)
  - Background information may encode bias, no legal basis for using

Performance across program

We use Area under the ROC Curve (AUC) to measure performance

- Higher is better (50% is random guess, 100% is perfect accuracy)
- Probability that the model will rank a randomly chosen positive instance higher than a randomly chosen negative instance

Insights

- ML outperforms current admission (quota 1)
- Transformer outperforms traditional ML:
  Gain equal to gain from protected attr.

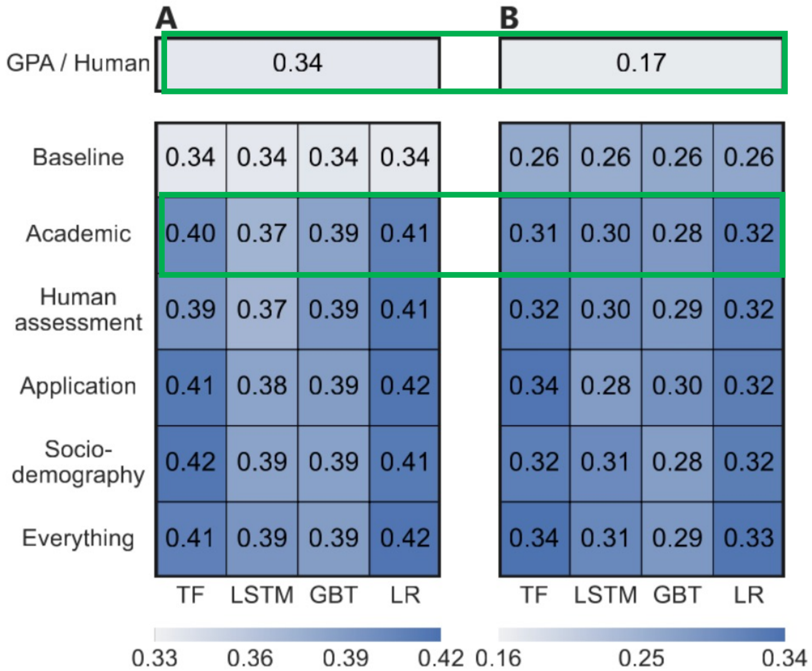
	LSTM	Gradient boost.	Logistic reg.	Transformer
GPA baseline	63.2	64.6	64.4	63.8
Academic	67.4	68.3	68.5	69.6
Application	68.0	69.2	69.3	70.5
Human assessment	67.5	68.8	68.8	69.8
Socio-demographic	67.1	68.8	68.6	68.9
Everything	68.3	69.1	69.6	70.3

63 67 71
AUC

Performance on other outcomes

Are predictions correlated with another outcome, e.g., grades obtained?

Quota 1 (GPA) *Quota 2 (Human assess)*

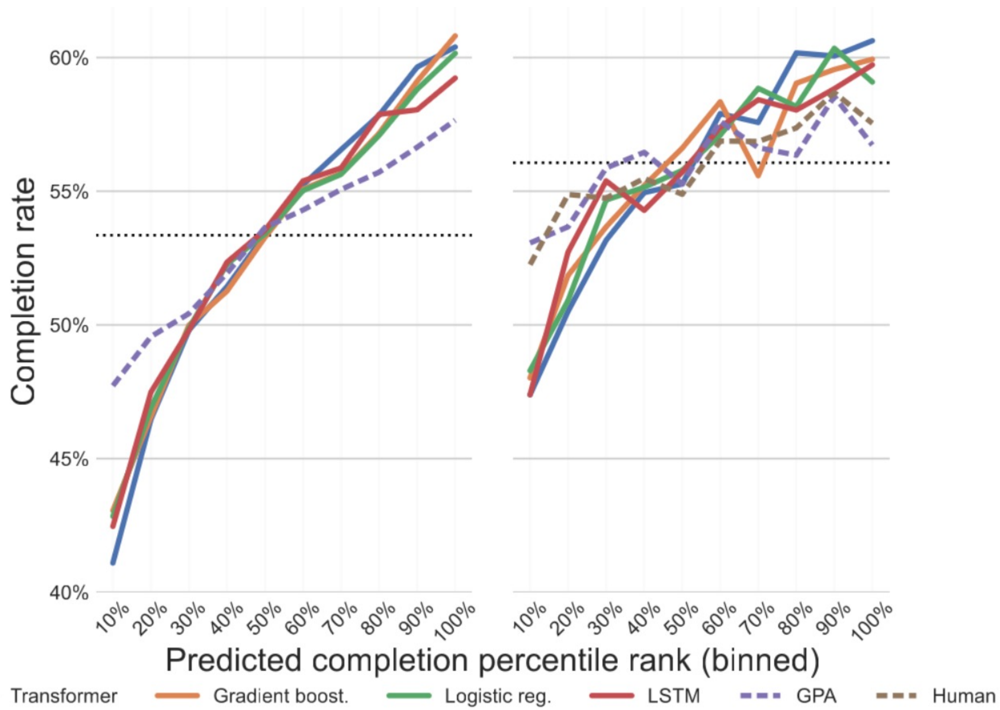


Performance within program

Rank all students based on predicted completion

- Ranking is done within each institution (split by by decile)

Quota 1 (GPA) *Quota 2 (Human assess)*



Hypothetical admission scenario

New DK policy from 2023: Cut 10% of intake going forward

Can evaluate how to target students using admission entry

- Ignore substitution effects – consider only actual program

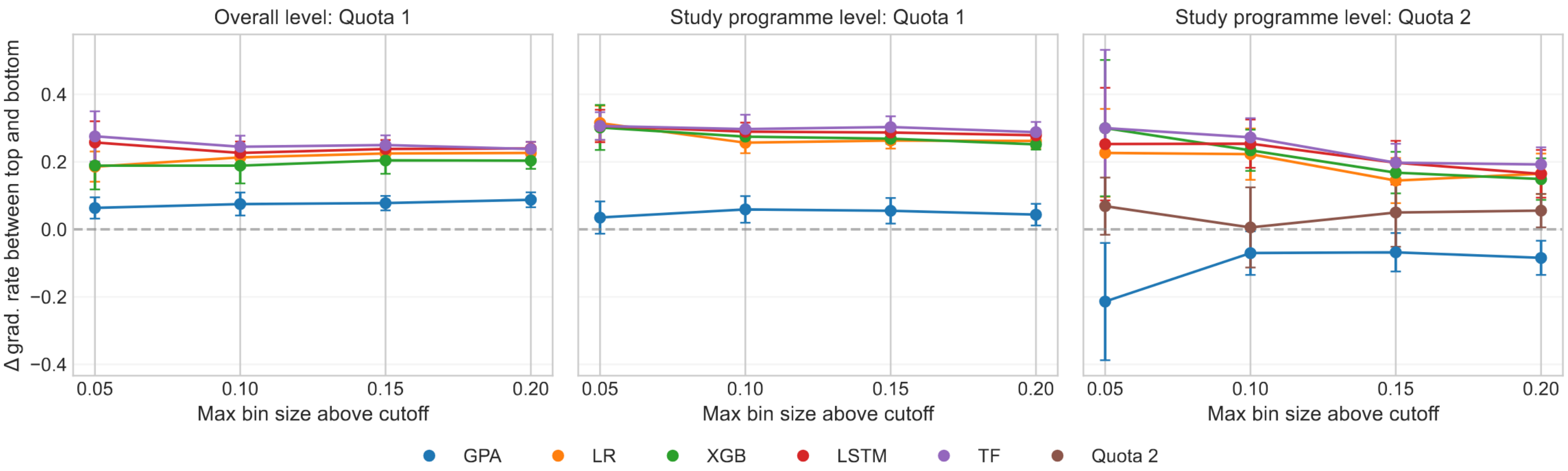
- Within program metrics are sufficient

Takeaways:

Better at identifying at-risk students than

- Current systems, both grade-based (GPA) and faculty screening
- Traditional machine learning methods

Marginal intake (new)



Algorithmic fairness

Trust in public policy depends on impartial treatment.

- Goal: Measure bias and discrimination in policies – how?
- Two common goals
  - Sufficient information: given a prediction (such as dropout risk), the likelihood of that prediction being correct should be equal for all groups
  - Equal opportunities: equal rate of admission chances across groups

- Problem – inherent tradeoff in these two measures (Kleinberg et al., 2017)
  - Alternative measure – difference in ROC-curve

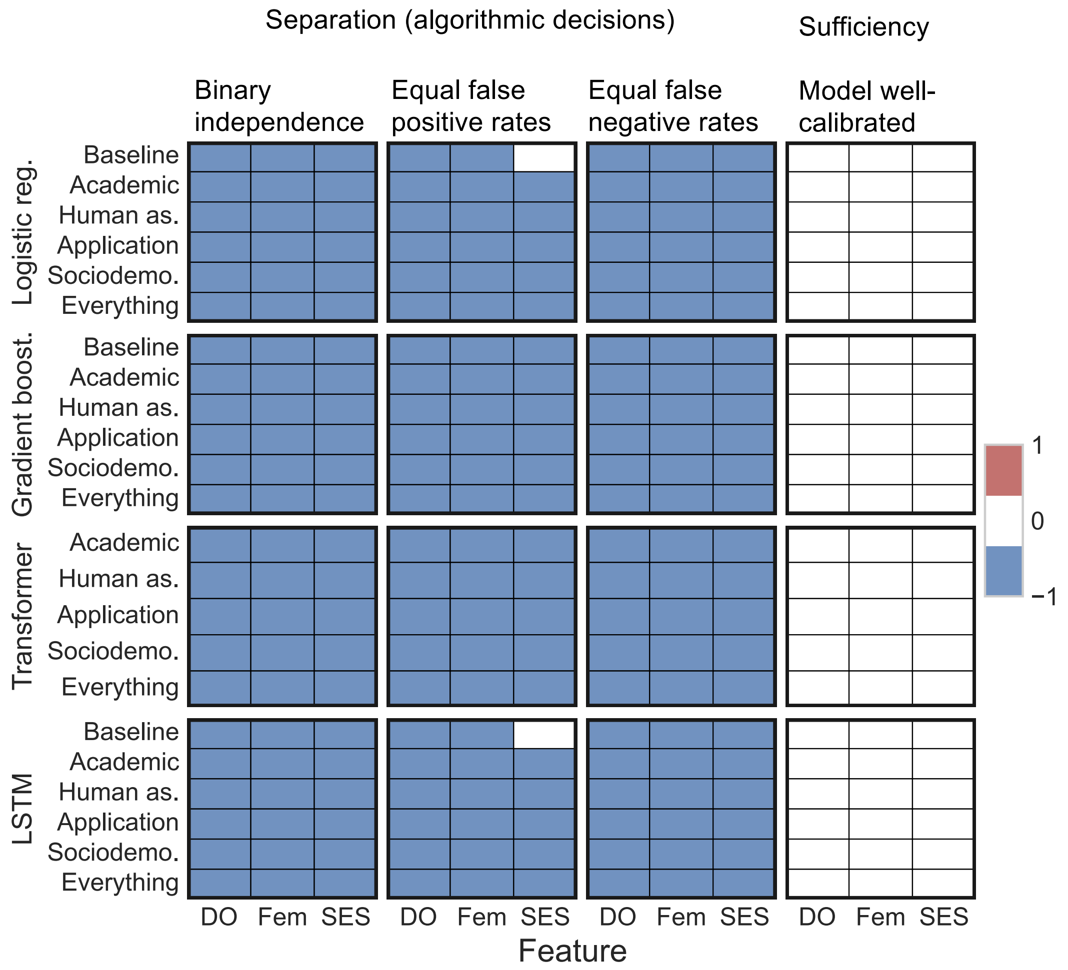
Algorithmic fairness

If the sensitive variables have predictive powers, the models have encoded it

- Across all models
- Across a wide variety of sensitive attributes

Findings

- Groups are not treated equally (*separation*)
- No fairness through unawareness (*sufficiency*)



Fairness compared to current systems

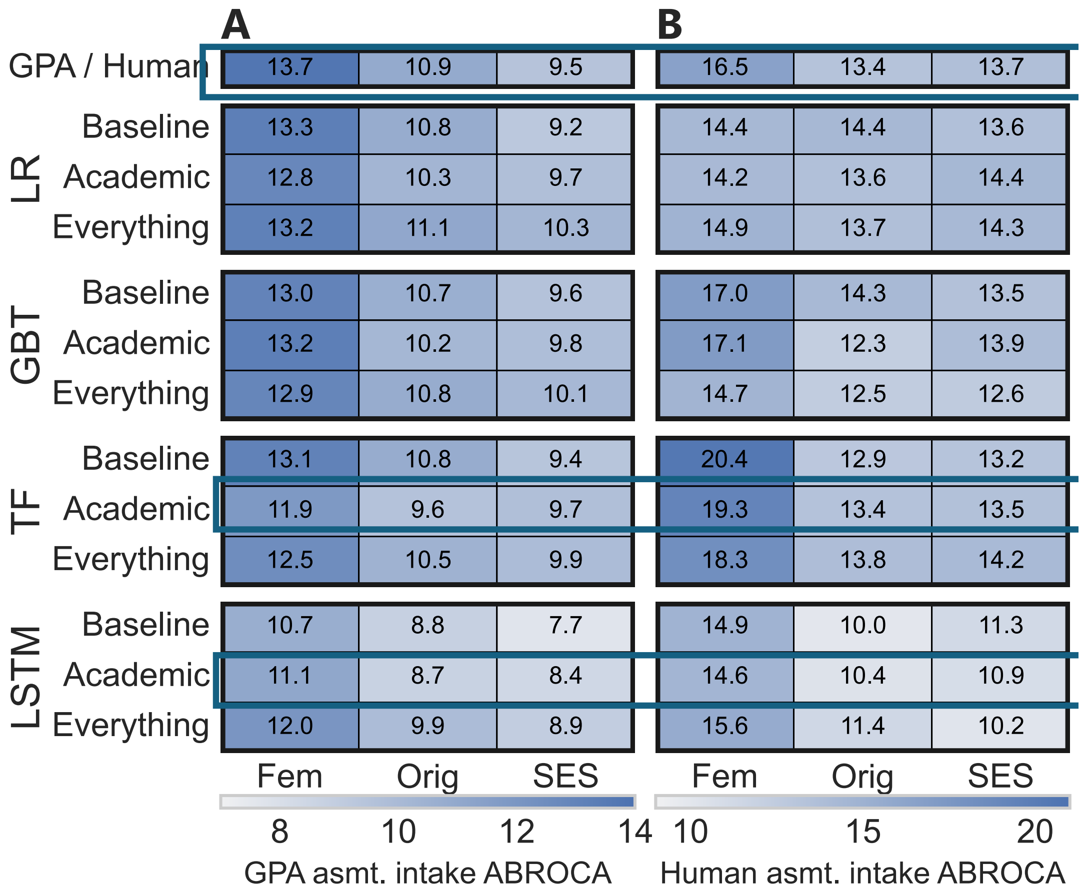
Compare models and current systems

- Use that each institution has a ranking of students used for admission

Fairness computed using ABROCA

- ABROCA measures difference in performance
  - ‘How good are the systems at ranking applicants’ across sensitive attributes
- Lower values indicate a more fair model

Transformer and LSTM better than Q1



Questions

- What happens if algorithm is used to re-rank students? How can we do counterfactual assessment?
  - What happens to other programs?
  - Would students change their college application behavior?

Grimon & Mills (2025)

- What: Prediction-guided child protection
  - Evaluate causal impact of deploying how algorithmic risk prediction tools
- How: Two key outcomes
  - efficiency (accuracy and workload in screening decisions)
  - equity (racial, gender, and socioeconomic disparities)
- Where: Large county in Colorado

Empirical Setting

- Review of reports of child abuse or neglect (“referrals”)
  - Daily review by multi-person CPS decision teams (~170 workers, 3–4 teams/day).
  - Each team decides whether to “screen in” (investigate) or “screen out” (no investigation).
  - About 30% of referrals are screened in.

Deployed algorithm

- Algorithm Tool:
  - Predicts the probability that a child will enter foster care within two years—a proxy for serious maltreatment risk.
  - Uses a LASSO-regularized logistic regression model trained on historical CPS data from the county and other Colorado counties.
  - Alternative models (Random Forests, XGBoost) were tested but not used due to computational efficiency.

Data sources

- Administrative**:** demographic and case-level data on all referrals during the trial.
- Hospital records**:** emergency room and inpatient visits (January 2020–June 2022, statewide)
  - *> objective measures of child harm post-referral*.
- Algorithm implementation**:** randomization logs and predicted risk scores for all referrals.

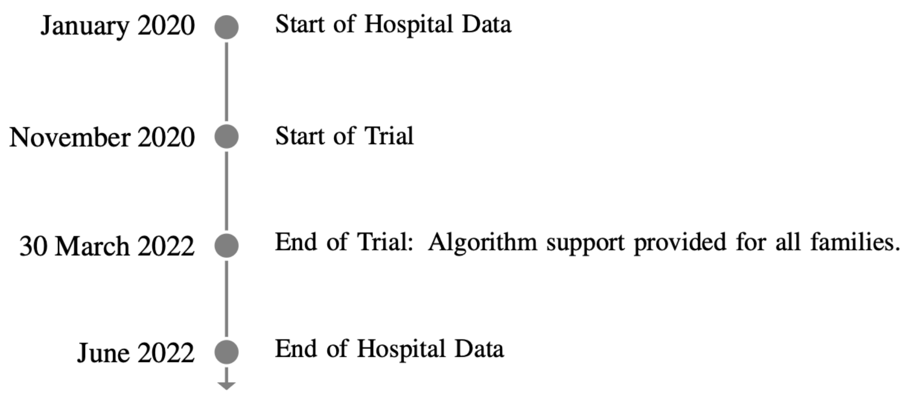
Implementation details

- Trial duration: 17 months (Nov 2020–Mar 2022).
- Sample size: 4,681 children (2,832 referrals) randomized.

Rollout and access

- Trial duration: 17 months (Nov 2020–Mar 2022).
- Sample size: 4,681 children (2,832 referrals) randomized.
- Unit of randomization:
  - the household (mother) — all children in a home share the same treatment status.
  - > avoids contamination from re-referrals of the same family
  - > some implementation issues

Figure B1: Data and Trial Timeline



Workflow integration and usage

- Treatment arms:
  - Intervention: display algorithm scores to team in the decision interface.
    - > risk scores for all children displayed
  - Control: CPS team did not see the scores (standard procedure).
    - > interface omitted all scores
  - Follow-up: Children retained their treatment status across re-referrals
- Exposure and usage
  - Tracking of whether workers saw algorithm scores
  - In ~73% of intervention cases wrote down scores (validated via text logs).
  - Consistent across risk levels.

Measured model

- Empirical model

$$Y_i = \delta_0 + \delta_1 \text{AlgoAccess}_{r(\mathrm{i})} + \gamma_{s(\mathrm{i})} + \varepsilon_i$$

- Y_i: screen-in, hospitalization
- AlgoAccess: binary variable of access, determined at first referral, $r(\mathrm{i})$
- γ: dummy variable for each sibling group size, $s(\mathrm{i})$

Impact on child health (hospital proxy)

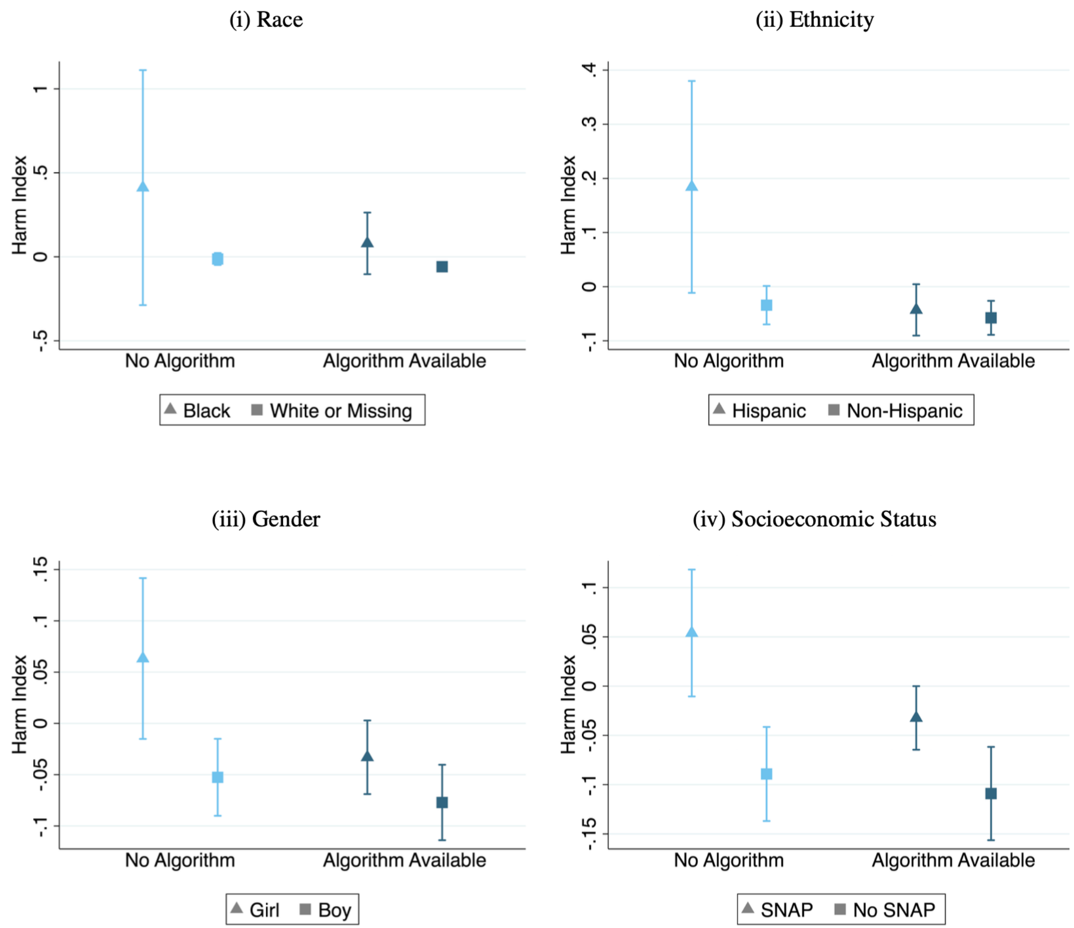
Table 2: Access to the Algorithm Tool Reduced Child Harm

Hospital Encounters after Randomization

	(1) Harm Index	(2) High Priority	(3) Injury	(4) Avoidable ER	(5) Maltreatment ICD	(6) Intentional Injury	(7) Top 1% in Harm Index
Algorithm Available	-0.061**	-0.10*	-0.060**	-0.020	-0.0014	-0.0072	-0.011***
	(0.030)	(0.056)	(0.024)	(0.022)	(0.0043)	(0.0047)	(0.0037)
Randomization Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Control Mean	0.000	0.660	0.210	0.168	0.013	0.020	0.016
Effect (%)		-16	-29	-12	-11	-36	-67
Power	0.81	0.80	0.82	0.48	0.18	0.48	0.86
Observations	3,431	3,431	3,431	3,431	3,431	3,431	3,431

Impact heterogeneity

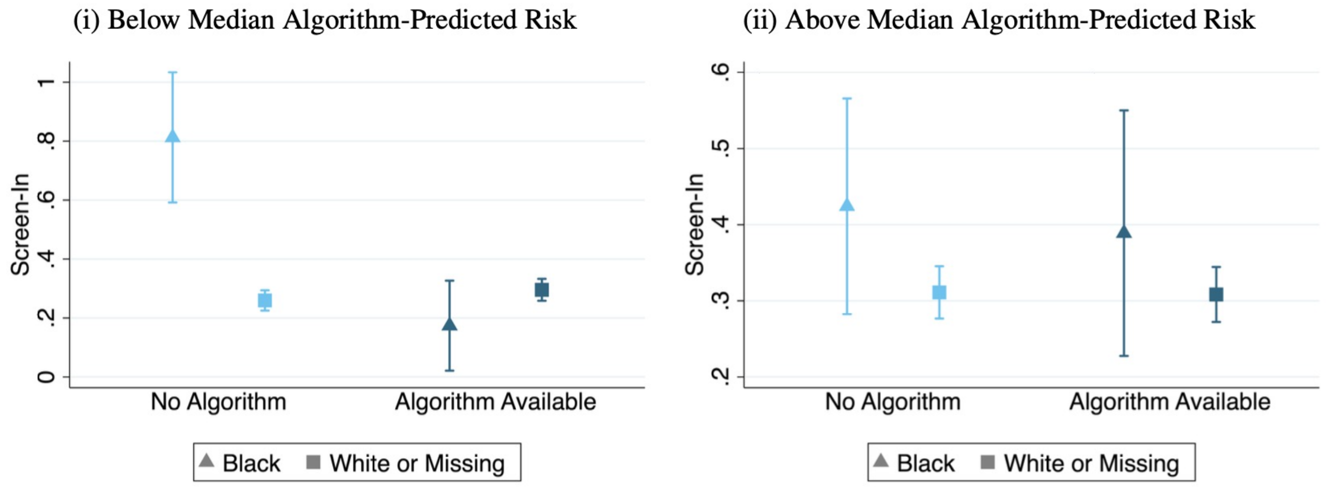
Figure 1: Algorithm Tool Reduced Harm the Most for Historically Disadvantaged Groups



Impact on initial screening

- Reduces disparities

Figure 2: Algorithm Tool Reduced Racial Disparities in CPS Surveillance for Low-Risk-Score Children



Notes: These figures present rates of CPS screen-in (investigation) by race and tool availability. Above and below-median predicted risk are defined using the algorithm referral risk score. Figures define race as recorded by Child Protective Services. 90% confidence intervals are reported and randomization controls are included throughout.

Exercise (3 min)

- Search for the Danish case by Gladsaxe municipality
- What happened with the algorithm?
- What would you recommend to policy-makers in Denmark?

LLM-powered decisions

Gen AI at work

Brynjolfsson, Li & Raymond (2025, QJE)

- Employ a quasi-experimental research design to measure impact of how a generative AI assistant

- Main outcomes: 1) worker productivity, 2) experience in a real-world workplace.

Implementation details

- The study examines a Fortune 500 company providing chat-based customer support for business software.
  - Number of agents: 5,172
  - Number of teams: 133

- A GPT-3–based AI assistant was introduced to help customer agents by offering real-time response suggestions and technical documentation links during chats.
  - Prior to launch of ChatGPT!

- The AI was designed to augment (not replace) workers, with agents free to ignore or edit suggestions.

Research design - event study

- The AI system was rolled out gradually (staggered adoption) across individuals between late 2020 and early 2021.
  - The staggered rollout enables a difference-in-differences (DiD) design comparing the same agents before and after AI access, while using non-treated agents as controls.
  - Can leverage modern event-study approaches

Research design - quasi-random exposure

- Key assumption in event study: quasi-random exposure

- Variation in adoption timing arose from limited training capacity and budget, creating quasi-random exposure across agents and teams.
  - Quasi-random scheduling: Managers allocated agents across sessions mainly to minimize customer-service disruptions, not based on performance or other productivity measures.
  - Constraints:
    - Training capacity: Only two trainers could conduct onboarding sessions, limiting the number of agents trained each week.
    - Budget: The firm had a fixed budget for how many agents could get access, and new slots opened only when AI-trained agents left.

Productivity effect

- Flat pre-trend (prior to exposure) indicates no selection effect

- Estimated effect of 0.3 cases per hour one month after in more robust model
  - ~ 15 percent

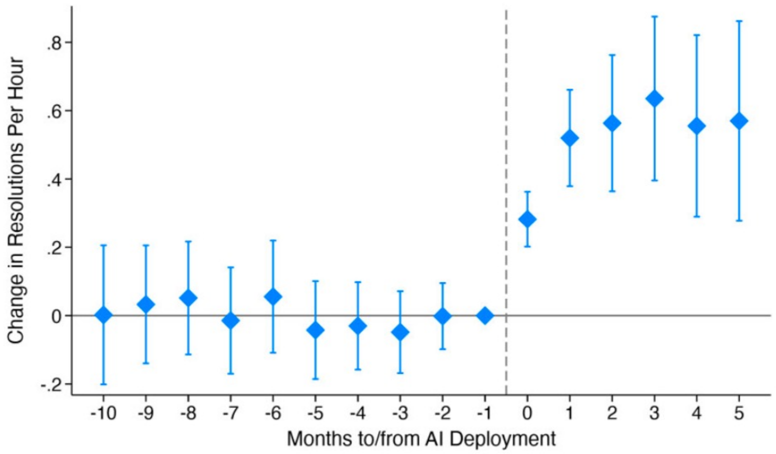


FIGURE II

TABLE II

MAIN EFFECTS: PRODUCTIVITY (RESOLUTIONS PER HOUR)

Variables	Resolutions/hour (1)	Resolutions/hour (2)	Resolutions/hour (3)
Post AI × Ever treated	0.469***	0.371***	0.301***
	(0.0325)	(0.0318)	(0.0329)
Ever treated	0.110**		
	(0.0440)		
Observations	13,192	12,295	12,295
R-squared	0.249	0.562	0.575
Year month FE	Yes	Yes	Yes
Location FE	Yes	Yes	Yes
Agent FE	—	Yes	Yes
Agent tenure FE	—	—	Yes
DV mean	2.123	2.176	2.176

Notes. This table presents the results of difference-in-difference regressions estimating the effect of AI model deployment on our main measure of productivity, resolutions per hour, the number of technical support problems resolved by an agent per hour (resolutions/hour). Post AI × Ever treated captures the impact of AI model deployment on resolutions per hour. Column (1) includes agent geographic location and year-by-month fixed effects. Columns (2) and (3) include agent-level fixed effects, and column (3), our preferred specification described by equation (1), also includes fixed effects that control for months of agent tenure. Observations for this regression are at the agent-month level and all standard errors are clustered at the agent level. Section IV.A describes the AI rollout procedure. Robust standard errors are in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$.

More on productivity

- What are the overall effects for an agent?
- Harm for quality?

Average Handling Time (per case, minutes)

Main Effects: Additional Outco

Variables	AHT (1)	Chats/hour (2)
Post AI × Ever treated	−3.746***	0.365***
	(0.369)	(0.0345)
Observations	21,839	21,839
R-squared	0.591	0.563
Year month FE	Yes	Yes
Agent FE	Yes	Yes
Agent tenure FE	Yes	Yes
DV mean	40.64	2.559

Notes. This table presents the results of difference-in-difference regressions estimating the effect of AI model deployment on additional measures of productivity and agent performance. Post AI × Ever treated measures the impact of AI model deployment after deployment on treated agents for average handle time (AHT) in column (1); chats per hour, the number of chats an agent handles per hour in column (2); resolution rate, the share of technical support problems they can resolve in column (3); and net promoter score (NPS), an estimate of customer satisfaction in column (4). Our regression specification, equation (1), includes fixed effects for each agent, chat year-month, and agent months of tenure. Observations for this regression are at the agent-month level and all standard errors are clustered at the agent level. Section IV.A describes the AI rollout procedure. Robust standard errors are in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$.

Heterogeneous adoption

- Authors analyze agents' adherence
  - (degree of compliance with recommendation)

- Positive impact on productivity

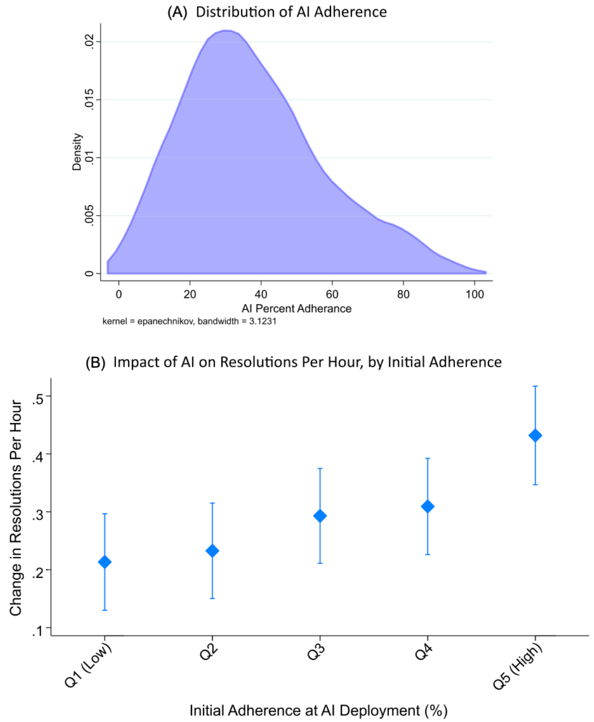


FIGURE VI

Heterogeneity of AI Impact, by AI Adherence

Panel A plots the distribution of AI adherence, averaged at the agent-month level, weighted by the log of the number of AI recommendations for that agent-month. Panel B shows the effect of AI assistance on resolutions per hour, by agents grouped by their initial adherence, defined as the share of AI recommendations they followed in the first month of treatment. The regression, outlined in Online Appendix J.C, is run at the agent-month level and includes fixed effects for agent, chat year-month, and agent tenure in months. Standard errors are clustered at the agent level. Results are in Online Appendix Table A.VII.

Adoption and learning

- Are gains permanent?
- Evidence from outages

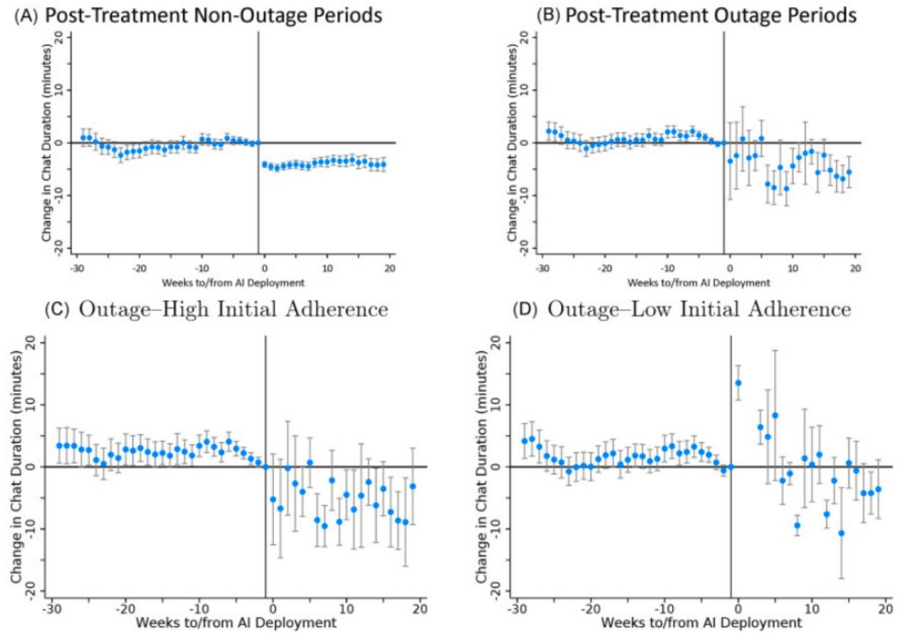


FIGURE VII

Chat Duration during AI System Outages

These figures plot event studies for the effect of AI system rollout of chat duration at the individual-chat level. Panel A restricts to posttreatment chats that do not occur during any period where there is a AI system outage. Panel B restricts to posttreatment chats that only occur during a large system outage. Panels C and D focus on outage only post-periods. Panel C restricts to only chats generated by ever-treated agents who with high initial AI adherence (top tercile), while Panel D restricts to agents with low initial adherence (bottom tercile). Agents who are never treated are excluded from this analysis. The regressions are run at the chat level with agent, year-month, and tenure fixed effects with standard errors clustered at the agent level. Regression results are reported in Online Appendix Table A.VIII.

Questions

- Do the effects generalize to other occupations?
- What do you think happened to the workers after 2022?
- What outcomes could be included in a more comprehensive study?

Takeaways from today

- Evidence of positive impact of AI on decisions
  - Admission screening
  - Workplace productivity (customer support)

- However..
  - Not for medical diagnosis when used by humans
  - Problems
    - Automation neglect
    - Signal dependence neglect

- Next time
  - Delegation
  - Causal ML intro